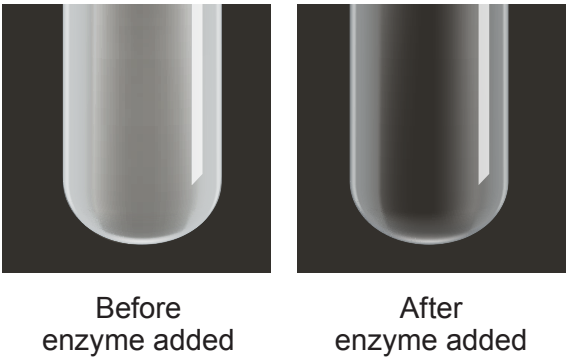


7. Dried milk powder and whey powder both contain protein. A weight lifter wanted to determine which would be more beneficial in his diet for his performance.

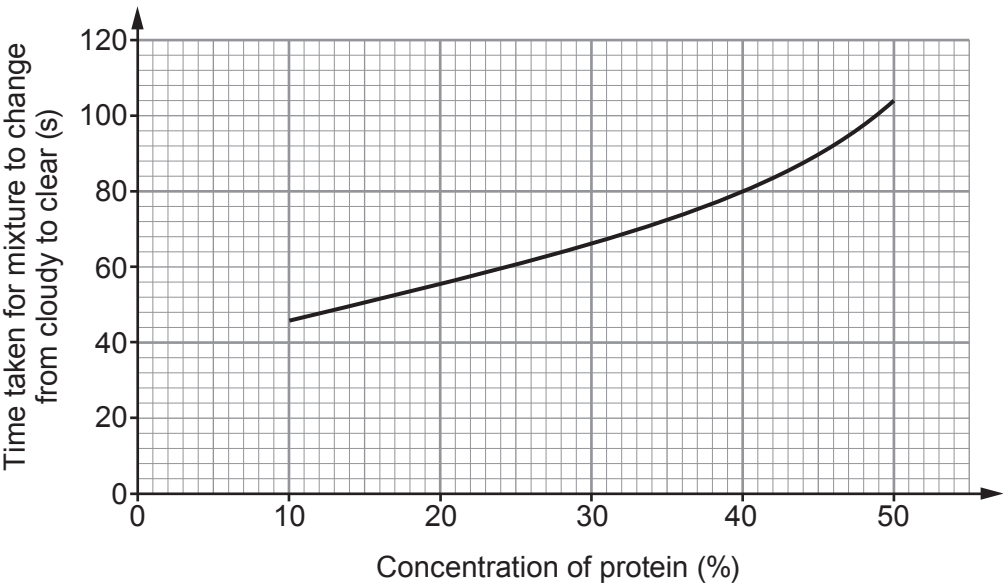
In order to do this, he first prepared equal volumes of known concentrations of protein for comparison. He then added 5 cm³ of 5% enzyme solution to each and recorded the time taken for the protein-enzyme mixture to go from cloudy to clear. **This colour change is shown in Image 7.1.**

Image 7.1



His results are shown in the **Graph 7.2.**

Graph 7.2



- (a) (i) State the independent and dependent variables. [2]
- Independent variable
- Dependent variable



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(ii) Name the type of enzyme which was added to the protein **and** explain the change in appearance from cloudy to clear. [2]

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(iii) State **one** source of uncertainty in the experimental method **and** suggest how it could be overcome. [2]

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(b) The weight lifter then carried out the experiment using a sample of dried milk powder followed by a sample of whey powder. The milk powder took 72 seconds to clear and the whey powder took 100 seconds to clear.

Use data from **Graph 7.2** to explain why the weight lifter concluded he should use whey powder rather than dried milk powder in his diet to improve his performance. [2]

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END OF PAPER

